# Jamie Lilly

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## Employment

2024 – Present **Postdoctoral Research Associate**, *Los Alamos National Laboratory*
Postdoctoral researcher in the Computational Physics and Methods group (CAI-2) and the Center for Nonlinear Studies (CNLS) at Los Alamos.

## Education

2019 – 2024 **Ph.D., Mathematics**, *Oregon State University*
Advised by Robert Higdon.
Dissertation title: *Efficient Time-Stepping for the Shallow Water Equations on Unstructured Grids*.

2019 – 2021 **M.S., Mathematics**, *Oregon State University*
Advised by Elaine Cozzi.

2016 – 2019 **B.S., Mathematics**, *Oregon State University*
Major in mathematics, minor in computer science.

2014 – 2016 **Associate of Arts Oregon Transfer**, *Portland Community College*

## Publications


[1] Jamie Lilly, Derek DeSantis, and Mark R. Petersen. "Viability of Tensor Train Methods for Geophysical Fluid Dynamics". In: *Journal of Computational Physics* 569 (Jan. 2027), p. 115465. ISSN: 0021-9991. DOI: 10.1016/j.jcp.2026.115465.

[2] M. Engin Danis, Duc P. Truong, Derek DeSantis, Jamie Lilly, Mark R. Petersen, Kim Ø Rasmussen, and Boian S. Alexandrov. "High-Order Tensor-Train Finite Volume Methods for Shallow Water Equations". In: *Monthly Weather Review* (June 2025). ISSN: 1520-0493, 0027-0644. DOI: 10.1175/MWR-D-24-0165.1.

[3] Jamie M. Lilly, Giacomo Capodaglio, Darren Engwirda, Robert L. Higdon, and Mark R. Petersen. "Local Time-Stepping for the Shallow Water Equations Using CFL Optimized Forward-Backward Runge-Kutta Schemes". In: *Journal of Computational Physics* 520 (Jan. 2025), p. 113511. ISSN: 0021-9991. DOI: 10.1016/j.jcp.2024.113511.

[4] Jamie M. Lilly, Darren Engwirda, Giacomo Capodaglio, Robert L. Higdon, and Mark R. Petersen. "CFL Optimized Forward–Backward Runge–Kutta Schemes for the Shallow-Water Equations". In: *Monthly Weather Review* 151.12 (Dec. 2023), pp. 3191–3208. ISSN: 1520-0493, 0027-0644. DOI: 10.1175/MWR-D-23-0113.1.

[5] Jamie M. Lilly, Giacomo Capodaglio, Mark R. Petersen, Steven R. Brus, Darren Engwirda, and Robert L. Higdon. "Storm Surge Modeling as an Application of Local Time-Stepping in MPAS-Ocean". In: *Journal of Advances in Modeling Earth Systems* 15.1 (Jan. 2023), e2022MS003327. ISSN: 1942-2466. DOI: 10.1029/2022MS003327.

[6] Jim Brown, Beren Gunsolus, Jamie Lilly, and Felice Manganiello. "Hilbert modular forms and codes over $\mathbb{F}_{p^2}$". In: *Finite Fields and Their Applications* 67 (2020), p. 101731. ISSN: 1071-5797. DOI: 10.1016/j.ffa.2020.101731.

## Grants

FY 2026 **Laboratory Directed Research and Development – Innovative Research**, *Los Alamos National Laboratory*
Role: Principle Investigator.
Project title: *Neural Differential Equations for Turbulent Mixing in the Ocean*.

FY 2025 **Laboratory Directed Research and Development – Rapid Response**, *Los Alamos National Laboratory*
Role: Principle Investigator.
Project title: *Sparse Tensor Trains for Geophysical Fluids on Unstructured Grids*.

## Research Programs

Summer 2023 **National Science Foundation Mathematical Sciences Graduate Internship**, *Los Alamos National Laboratory*
Developed CFL efficient local time-stepping schemes for shallow water models.

2022 – 2023 **Department of Energy Science Graduate Student Research Program**, *Los Alamos National Laboratory*
Investigated the performance of local time-stepping schemes in MPAS-Ocean to increase computational efficiency. Developed CFL optimized Runge-Kutta schemes for the shallow water equations.

Summer 2021 **Parallel Computing Student Research Internship**, *Los Alamos National Laboratory*
Designed, built, and ran performance experiments for local time-stepping schemes for MPAS-Ocean using HPC systems.

2018 – 2019 **Undergraduate Honors Thesis**, *Oregon State University*
An independent study of the gauge integral, including a general overview and major convergence theorems.

Summer 2018 **Research Experience for Undergraduates**, *Clemson University*
Developed a method to construct lattices from linear codes from certain finite fields using tools from algebraic number theory.

## Scholarly Presentations

### Conference Talks

July 24, 2023 U.S. National Congress on Computational Mechanics
*Storm Surge Modeling as an Application of Local Time-Stepping in MPAS-Ocean*

March 1, 2022 American Geophysical Union Ocean Sciences Meeting
*Speeding Up Ocean Simulations with Local Time-Stepping*

### Seminar Talks

April 5, 2025 Center for Nonlinear Studies Postdoc Seminar
*Viability of Tensor Train Methods for Geophysical Fluid Dynamics*

May 5, 2025 Center for Nonlinear Studies Colloquium
*Towards Tensor Decompositions for Geophysical Fluids*

October 23, 2024 LANL Climate, Ocean, and Sea Ice Modeling Seminar
*Efficient Time-Stepping for the Shallow Water Equations on Unstructured Grids*

April 12, 2024 OSU Applied Math and Computation Seminar
*CFL Optimized Local Time-Stepping for the Shallow Water Equations*

August 2, 2023 LANL Climate, Ocean, and Sea Ice Modeling Seminar
*CFL Optimized Forward-Backward Runge-Kutta Schemes for the Shallow Water Equations*

March 3, 2023 — OSU Applied Math and Computation Seminar
*Storm Surge Modeling as an Application of Local Time-Stepping in MPAS-O*

June 8, 2022 — LANL Climate, Ocean, and Sea Ice Modeling Seminar
*Storm Surge Modeling as an Application of Local Time-Stepping in MPAS-O*

August 11, 2021 — LANL Climate, Ocean, and Sea Ice Modeling Seminar
*Speeding Up Ocean Simulations with Local Time-Stepping*

July 8, 2018 — Multi-REU Mock Conference
*Codes, Lattices, and Modular Forms*

### Poster Sessions

June 4, 2025 — Algorithms For Multiphysics Models In The Post-Moore's Law Era
*Towards Tensor Decompositions for Geophysical Fluids*

February 20, 2024 — American Geophysical Union Ocean Sciences Meeting
*CFL Optimized Forward-Backward Runge-Kutta Schemes for the Shallow Water Equations*

August 9, 2023 — Los Alamos Workshop on Time Integration for Multiphysics
*CFL Optimized Forward-Backward Runge-Kutta Schemes for the Shallow Water Equations*

### Professional Service

February 18, 2024 — Session co-organizer at American Geophysical Union Ocean Sciences Meeting
*Numerical Methods for Computational Oceanography*

## Awards and Honors

May 2024 — Graduate Student Excellence Award

June 2023 — Graduate Student Academic Achievement Award

June 2021 — Graduate Student Outstanding Performance in Coursework Award

May 2018 — Botond Gabor Eross Math Memorial Scholarship

April 2017 & 2018 — Drucilla Shepard Smith Award

## Other Experience

### Teaching Assistantships

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| Fall 2019 | Multivariable Calculus | Winter 2020 | College Algebra |
| Spring 2020 | Differential Calculus | Fall 2020 | College Algebra |
| Winter 2021 | Integral Calculus | Spring 2021 | Integral Calculus |
| Fall 2021 | Multivariable Calculus | Winter 2023 | Multivariable Calculus |
| Spring 2023 | Multivariable Calculus | Fall 2024 | Mulitvariable Calculus |
| Winter 2024 | Numerical ODEs | Spring 2024 | Linear Algebra |

### Other Employment

2019 – 2024 — **Graduate Student**, *Oregon State University*

Funded by OSU Graduate Teaching Assistantship, Department of Energy Office of Science Graduate Student Research Award, Los Alamos National Laboratory Parallel Computing Summer Research Internship, and National Science Foundation Mathematical Sciences Graduate Internship.

Summer 2019 & 2020 **Software Engineering Intern**, *Engineering Design Team*
Wrote computer vision applications on Nvidia Jetson machines using PyTorch and Tensorflow Keras, and wrote applications that perform inference on images pulled from propietary EDT hardware. Used Amazon Web Service virtual machines and Tensorflow to build and train a ML model to classify radio signals.

Winter 2018 **Grader**, *OSU Math Department*
Graded weekly homework assignments for Introduction to Modern Algebra.

## Service

October 2020 OSU math department promotion and tenure student subcommittee.

April 2019 OSU math department faculty awards selection committee.

November 2017 OSU math department promotion and tenure student subcommittee.